SC3102 Part 2 Cheat Sheet · Side 1: signals, systems, DFT, DTFT

1. Toolkit

$$\Sigma_{n=0}^{\infty} q^n = \frac{1}{1-q}\ (|q|<1) \quad \Sigma_{n=1}^{\infty} q^n = \frac{q}{1-q} \quad \Sigma_{n=0}^{N-1} q^n = \frac{1-q^N}{1-q}\ (q\neq 1)$$

- **Euler:** $\cos\theta = (e^{j\theta} + e^{-j\theta})/2$, $\sin\theta = (e^{j\theta} - e^{-j\theta})/(2j)$. $1/j = -j$, $e^{j\pi/2} = j$, $e^{j\pi} = -1$.
- **Sampling:** $x[n] = x_c(nT_s)$, $T_s = 1/f_s$. $\Omega = \omega T_s = 2\pi f/f_s$ (rad/sample).
- **Period of $\cos(\Omega_0 n)$:** write $\Omega_0/2\pi = m/N$ in lowest terms → $N_0 = N$ (the denominator).
  - $\sin(\pi n/12)$: $1/24$ → $N_0 = 24$. $\cos(3.5\pi n)$: $7/4$ → $N_0 = 4$.
  - $\Omega_0/2\pi$ irrational (e.g. $\cos(n)$) → not periodic.
- **Sum of periodic signals:** $N_0 = \text{LCM}(N_1, N_2)$.
- **Nonzero periodic signal is never absolutely summable** ($\Sigma|x|$ adds the same block forever).
- **Aliasing:** Ω, $\Omega + 2\pi k$ and (for cos) $-\Omega$ give identical samples. In Hz: f, $f + k\cdot f_s$, $f_s - f$ look the same.
  - e.g. $\cos(3.5\pi n) = \cos(-0.5\pi n) = \cos(0.5\pi n)$.
  - Unique only if $0 \le \Omega \le \pi$, i.e. $f < f_s/2$. Nyquist: $f_s > 2f_{max}$.

2. Systems and convolution

Property	Means	Quick test / red flags
Linear	$T(ax+bz) = a\cdot Tx + b\cdot Tz$	Double the input: does the output double? Fails with x^2, $x[n]\cdot x[n-1]$, + constant
Time-inv.	input shifted by r → output shifted by r	Fails if n appears outside x[·]: $n\cdot x[n]$, $x[2n]$, $x[-n]$
Causal	y[n] uses only x[n], x[n−1], …	Fails with any x[n+1] (future input)
BIBO stable	$\|x\| \le M \to \|y\| \le C$	Bound term by term: $\|x[n]+x[n-1]+x[n]x[n-1]\| \le 2M + M^2$

- **LTI:** $h = T(\delta)$; $y = x * h$, $(x * h)[n] = \Sigma_m x[m]\cdot h[n-m]$.
- **LTI causal** ⇔ $h[n] = 0$ for $n < 0$. **LTI stable** ⇔ $\Sigma|h[n]| < \infty$.
- **Convolution facts:** $x * \delta[n-k] = x[n-k]$. $x * u$ = running sum $\Sigma_{m\le n} x[m]$. $u * u = (n+1)u[n]$.
- **Step → impulse response:** $s = T(u) \Rightarrow h[n] = s[n] - s[n-1]$ (because $\delta[n] = u[n] - u[n-1]$).
- **Length:** linear conv of L_x and L_h samples has $L_x + L_h - 1$ samples.
- **FIR:** no y on the right; h = the coefficients; all poles at z = 0; always stable.
  IIR: feedback (y[n−k] on the right); h never ends; stable only if all poles inside |z| = 1.

3. DFT and DTFS

$$X[k] = \Sigma_{n=0}^{N-1} x[n]\, e^{-j2\pi kn/N} \qquad x[n] = \frac{1}{N}\Sigma_{k=0}^{N-1} X[k]\, e^{+j2\pi kn/N}$$

- **Bin k** ↔ $\Omega_k = 2\pi k/N$ ↔ $f_k = k\cdot f_s/N$.
- **N = 4:** X = (matrix)·x. Synthesis x = ¼·(same matrix with $j \leftrightarrow -j$)·X.

Analysis row	n=0	n=1	n=2	n=3	Synthesis row (×¼)
k=0	1	1	1	1	1, 1, 1, 1
k=1	1	−j	−1	j	1, j, −1, −j
k=2	1	−1	1	−1	1, −1, 1, −1
k=3	1	j	−1	−j	1, −j, −1, j

- **Quick values:** $X[0] = \Sigma x[n]$. $x[0] = (1/N)\Sigma X[k]$. N = 4: $X[2] = x_0 - x_1 + x_2 - x_3$.
- **Real x:** $X[N-k] = X[k]^*$ (N = 4: X[3] = X[1]*); X[0] and X[N/2] are real. Rows 1 and 3 are conjugates.
- **Unknowns in x and X (T02 Q1d style):** fill in X[3] = X[1]* first, then use one synthesis row for a known x value.
- **DTFS:** same formulas, valid for all n only if N is a period (multiple of N_0); otherwise the repeated block ≠ the signal (leakage).
- **N_0 vs N:** N_0 = signal's period (calculate). N = DFT size (choose). No leakage ⇔ N is a multiple of N_0.
- **DTFS by Euler (no sum):** write x as a sum of $e^{j2\pi kn/N}$; then X[k] = N × (coefficient of $e^{j2\pi kn/N}$). Use $e^{-j2\pi n/N} = e^{j2\pi(N-1)n/N}$.
  - e.g. $\cos(2\pi n/8) = \tfrac{1}{2}e^{j2\pi n/8} + \tfrac{1}{2}e^{j2\pi\cdot 7n/8}$ → X[1] = X[7] = 8 × ½ = 4.
  - For sin: $1/(2j) = \tfrac{1}{2}e^{-j\pi/2}$.

4. Circular convolution and the DTFT

- **Multiplying DFTs gives circular convolution:** $\text{IDFT}_N\{X[k]\cdot Y[k]\} = x \circledast_N y$, not the linear one.
- **Wrap rule:** C[n] = L[n] + L[n+N] (+ L[n+2N] …): fold the tail of the linear result L back onto its start.
  - e.g. L = [−2, 1, 3, −6, 2, 8], N = 4 → C = [−2+2, 1+8, 3, −6] = [0, 9, 3, −6].
- **Circular = linear:** zero-pad both to $N \ge L_x + L_y - 1$.

$$X(\Omega) = \Sigma_{n=-\infty}^{\infty} x[n]\, e^{-j\Omega n} \qquad x[n] = \frac{1}{2\pi}\int_{2\pi} X(\Omega)\, e^{j\Omega n}\, d\Omega$$

- **Repeats every 2π** in Ω. **Exists if $\Sigma|x[n]| < \infty$**; u[n], $2^n u[n]$ and periodic signals fail.
- **Pairs ($|a| < 1$):** $a^n u[n] \leftrightarrow 1/(1 - a\cdot e^{-j\Omega}) \cdot a^{-n}u[-n-1] \leftrightarrow a\cdot e^{j\Omega}/(1 - a\cdot e^{j\Omega}) \cdot (n+1)a^n u[n] \leftrightarrow 1/(1 - a\cdot e^{-j\Omega})^2$ (= $x * x$).
- **Shift:** $x[n-k] \leftrightarrow e^{-j\Omega k}\cdot X(\Omega)$ (size same, only phase changes).
- **Convolution → product:** $Y(\Omega) = H(\Omega)\cdot X(\Omega)$. $H(\Omega)$ = DTFT of h = H(z) at $z = e^{j\Omega}$.
- **One frequency in → same frequency out:** $e^{j\Omega_0 n} \to H(\Omega_0)\cdot e^{j\Omega_0 n}$. |H| = gain, ∠H = phase shift.

9. Q&A notes (things I asked about)

- **Period rule, step by step:** $\Omega_0 N = 2\pi m$, where N = samples (a whole number) and m = full laps. Find the smallest m that makes N whole. $\Omega_0/2\pi = f/f_s$ = cycles per sample; periodic ⇔ rational. Reduce to lowest terms: bottom = N_0, top = cycles.
- **Simplify first:** the rule gives when the *walker* returns. It overcounts only if every sample is 0: $\sin(\pi n) = \sin(3\pi n) = 0$, which has period 1 (the rule says 2). A sign on Ω or $\pm 2\pi$ aliasing doesn't change N_0. $\cos(-\theta) = \cos\theta$; $\sin(-\theta) = -\sin\theta$.
- **$T_s \neq T_0 \neq N_0$:** $T_s = 1/f_s$ (sampler, s); $T_0 = 1/f$ (wave, s); N_0 (samples, whole number). $N_0\cdot T_s = m\cdot T_0$.
- **Causal, formally:** if x = z up to time N, then the outputs are equal up to N. In practice: every x[·] index ≤ n for every n. Traps: x[2n], x[−n]. $n\cdot x[n]$ is fine.
- **Linearity:** $T(a\cdot x_1 + b\cdot x_2) = a\cdot Tx_1 + b\cdot Tx_2$, where a, b scale *values* (x[an] is time scaling, a different thing). **Disprove** with 1 counterexample that switches on the suspect term: for $x[n]x[n-1]$, use $v = \delta[n] + \delta[n-1]$: T(v)[1] = 3, T(2v)[1] = 8 ≠ 6. **Prove** with algebra: product terms leave $(a^2-a)pq + (b^2-b)rs + ab(ps + rq)$, via $a^2pq = a\cdot pq + (a^2-a)pq$.
- **Convolution means:** $x = \Sigma x[m]\delta[n-m]$; each kick sends out $x[m]\cdot h[n-m]$; add the copies (LTI only). [2, 1] * [1, 2, 3] = [2, 5, 8, 3].
- **Running sum, h = u:** $y[n] = \Sigma_{m\le n} x[m]$; input u → (n+1)u[n]. Bounded in, unbounded out → unstable (pole at z = 1).
- **DFT as matrices:** $\omega = e^{j2\pi/N}$ (N = 4: ω = j). $G[n,k] = \omega^{nk}$. G^H = conjugate transpose (G is symmetric, so this just swaps $j \leftrightarrow -j$). $X = G^H x$ is the analysis sum (row k · x = X[k]); $x = (1/N)\cdot G\cdot X$ is the synthesis. It works because $G\cdot G^H = N\cdot I$. **Why −j:** multiplying by the opposite spin stops a matching walker ($j^n\cdot(-j)^n = 1$ each time, which sums to N), and any other frequency cancels to 0.
- **Signal or system?** Transform a signal → X: what it contains. Transform h → H: what the system does to each frequency. Y = H·X. ROC for a signal → which side, whether the DTFT exists; for a system → causal, stable.

SC3102 Part 2 Cheat Sheet · Side 2: Z-transform and difference equations

5. Z-transform and its pairs

$$X(z) = \Sigma_{n=-\infty}^{\infty} x[n]\, z^{-n}, \quad z = r \cdot e^{j\Omega}$$

- **Meaning:** $X(re^{j\Omega})$ = DTFT of the faded signal $r^{-n} \cdot x[n]$. $|z| = r$ = fading, $\angle z = \Omega$ = frequency. On $|z| = 1$: $X(e^{j\Omega})$ = DTFT.
- **Undo the fading:** if $W(\Omega) = X(re^{j\Omega})$ then $w[n] = r^{-n}x[n]$, so $x[n] = r^n \cdot w[n]$.
- **Tags:** finite list → $X(z) = \Sigma\, x[n]z^{-n}$. Coefficient of $z^{-k} = x[k]$ (future); of $z^{+k} = x[-k]$ (past).
- **Same thing, two forms:** $z/(z-p) = 1/(1 - p \cdot z^{-1})$.
- **DTFT inside X(z):** the Ω axis from $-\pi$ to π wraps round the unit circle ($\Omega = 0 \to z = 1$, $\pi/2 \to j$, $\pm\pi \to -1$). DTFT = 1 dial (Ω); Z = 2 dials (r, Ω). Building blocks are spirals $z^n = r^n e^{j\Omega n}$; a pole p means the signal contains p^n. e.g. $2^n u[n]$ faded with r = 4 gives 0.5^n (works for any r > 2, which is the ROC).

Signal x[n]	X(z)	ROC
$\delta[n-k]$	z^{-k}	all z (not 0 if k>0, not ∞ if k<0)
$u[n]$	$z/(z-1)$	$\|z\| > 1$
$-u[-n-1]$	$z/(z-1)$	$\|z\| < 1$
$p^n u[n]$	$z/(z-p)$	$\|z\| > \|p\|$
$-p^n u[-n-1]$	$z/(z-p)$	$\|z\| < \|p\|$
$p^{n-1} u[n-1]$	$1/(z-p)$	$\|z\| > \|p\|$
$-p^{n-1}u[-n]$ (optional)	$1/(z-p)$	$\|z\| < \|p\|$
$n \cdot p^n u[n]$	$pz/(z-p)^2$	$\|z\| > \|p\|$
$-n \cdot p^n u[-n-1]$ (optional)	$pz/(z-p)^2$	$\|z\| < \|p\|$
$(n+1) \cdot p^n u[n]$	$z^2/(z-p)^2$	$\|z\| > \|p\|$
$\rho^n \cos(\Omega_0 n) u[n]$	$z(z - \rho\cos\Omega_0) / (z^2 - 2\rho z \cdot \cos\Omega_0 + \rho^2)$	$\|z\| > \rho$
$\rho^n \sin(\Omega_0 n) u[n]$	$\rho z \cdot \sin\Omega_0 / (z^2 - 2\rho z \cdot \cos\Omega_0 + \rho^2)$	$\|z\| > \rho$

Special case $\Omega_0 = \pi/2$: $\rho^n \cos(\pi n/2) u[n] \leftrightarrow z^2/(z^2 + \rho^2)$; $\rho^n \sin(\pi n/2) u[n] \leftrightarrow \rho z/(z^2 + \rho^2)$; both $|z| > \rho$.

6. Properties and ROC rules

Property	Signal	X(z)	ROC (x: $r_1 < \|z\| < r_2$)
Linear	$a \cdot x + b \cdot y$	$a \cdot X + b \cdot Y$	contains the overlap
Delay	$x[n-k]$	$z^{-k} \cdot X(z)$	same (except maybe 0, ∞)
Time flip	$x[-n]$	$X(1/z)$	$1/r_2 < \|z\| < 1/r_1$
$\times a^n$	$a^n \cdot x[n]$	$X(z/a)$	$\|a\|r_1 < \|z\| < \|a\|r_2$
Convolution	$x * h$	$X(z) \cdot H(z)$	contains the overlap

- **Poles** = roots of the denominator (×); **zeros** = roots of the numerator (○). First make ONE fraction in +powers of z and cancel: zeros hide in sums ($z/(z-0.6) + z/(z-1.2) = 2z(z-0.9)/(\ldots)$). LCCDE: x-terms → zeros, y-terms → poles.
- **ROC is a ring** $r_1 < |z| < r_2$ with edges at poles; never contains a pole.
- **Which ring:** right-sided → outside the outermost pole · left-sided → inside the innermost pole · two-sided → a ring between poles (can be empty: $2^{|n|}$ needs $|z| > 2$ and $|z| < ½$ → no ROC) · finite length → all z except maybe 0 and ∞.
- **P poles at different radii** → P + 1 possible ROCs, each a different x[n].
- **Which way each pole runs:** pole inside the ring ($|p| \le$ inner edge) → right-sided $p^n u[n]$; pole outside ($|p| \ge$ outer edge) → left-sided $-p^n u[-n-1]$.
- **Stable (DTFT exists)** ⇔ unit circle $|z| = 1$ is in the ROC. A pole on $|z| = 1$ (e.g. z = 1) → no DTFT.
- **Causal** ⇔ ROC is outside the outermost pole **and** includes ∞ (numerator degree ≤ denominator degree in z).
- **Causal and stable** ⇔ every pole strictly inside the unit circle.

7. Inverting X(z)

1. **Factor the denominator** → poles. Mark the ROC's edges.
2. **Numerator has a factor z:** expand $X(z)/z = C/(z-p) + D/(z-q)$ → $X(z) = C \cdot z/(z-p) + D \cdot z/(z-q)$ → p^n-type terms.
3. **No factor z:** expand $X(z) = C/(z-p) + D/(z-q)$ → $p^{n-1}u[n-1]$-type terms.
4. **Cover-up:** C = [(z−p) × the fraction being expanded] at z = p.
5. **Choose each term's side from the ROC** (section 6), then write x[n].
6. **Special denominators:** repeated $(z-p)^2$ → $n \cdot p^n$ or $(n+1)p^n$ pairs · quadratic $z^2 - 2\rho z \cdot \cos\Omega_0 + \rho^2$ → cos/sin pairs (ρ^2 = constant term, then $\cos\Omega_0$ from the middle) · only z^2 appears → set $w = z^2$, do partial fractions in w.

Example: $X(z) = z/((z-½)(z-2))$, $|z| > 2$. $X/z = 1/((z-½)(z-2))$: $C = 1/(½-2) = -⅔$, $D = 1/(2-½) = ⅔$. Both poles inside the ring → right-sided: $x[n] = [-⅔(½)^n + ⅔ \cdot 2^n]u[n]$.

Series expansion (first few samples, right-sided): long-divide in powers of z^{-1}; x[n] = coefficient of z^{-n}. e.g. $(1 + z^{-1})/(1 - ½z^{-1}) = 1 + 1.5z^{-1} + 0.75z^{-2} + 0.375z^{-3} + \ldots$ → x[0..3] = 1, 1.5, 0.75, 0.375.

Checks: causal x: $x[0] = \lim_{z\to\infty} X(z)$. Plug in n = 0, 1 and compare.

8. Difference equations (LCCDEs)

No initial conditions (bilateral, LTI) → H(z)

1. Transform every term: $y[n-k] \to z^{-k}Y$, $x[n-k] \to z^{-k}X$.
2. $H(z) = Y/X = (b_0 + b_1z^{-1} + \ldots)/(1 + a_1z^{-1} + \ldots)$. Multiply top and bottom by the highest power of z to read the poles.
3. One H(z), several ROCs → several systems; use section 6 for causal/stable. Leftover positive power of z on top (from x[n+1]) → not causal for any ROC.
4. $H = z^{-k} \cdot G \Rightarrow h[n] = g[n-k]$: invert G, then shift.

With initial conditions (unilateral, n ≥ 0): shift rules

$$Z_+\{y[n-1]\} = z^{-1}Y_+ + y[-1] \qquad Z_+\{y[n-2]\} = z^{-2}Y_+ + z^{-1}y[-1] + y[-2]$$

An input starting at n = 0 has X_+ = its usual right-sided pair.

Recipe

1. Transform both sides with the shift rules.
2. Collect: $Y_+ \cdot (1 + a_1z^{-1} + \ldots) = X_+ -$ (initial-condition terms).
3. **Zero-input:** initial-condition terms ÷ $(1 + a_1z^{-1} + \ldots)$. **Zero-state:** $H \cdot X_+$.
4. Partial fractions ($w = z^2$ if only z^2 appears), invert with right-sided pairs (n ≥ 0).
5. **Complete:** $y = y_{zi} + y_{zs}$.
6. **Transient** = terms that die out (system poles inside $|z| = 1$). **Steady-state** = terms that last (input poles on $|z| = 1$).
7. **Remove the transient:** set $y[n] = y_{ss}[n]$ for all n; plug into the equation at n = 0 and n = 1; solve for y[−1], y[−2].
8. **Check:** compute y[0], y[1] straight from the equation. Initial value: $y[0] = \lim_{z\to\infty} Y_+(z)$. *(optional) Final value: $y[\infty] = \lim_{z\to 1}(1 - z^{-1})Y_+(z)$, only if all poles inside the unit circle except at most one at z = 1.*

A system is LTI only if its initial conditions are zero.

9. Q&A notes (continued)

Why poles are ROC edges: pole p ⇔ signal has p^n (×p each step); $\Sigma(p/z)^n$ needs $|z| > |p|$ → edge $|z| = |p|$. 1st-order LCCDE: pole = number before y[n−1] ($y = 0.5y[n-1] + x \to z/(z-0.5)$). Zero: input z^n → output 0. ROC = adds up *with* weight z^{-n}; stable = with no weight ($|z| = 1$). A blowing-up h can still have an ROC (Lab 4 Region 0).